

1.3 Homework: Measurement quiz review

Reference (provided). average speed = $\frac{\text{distance}}{\text{time}}$

1 in. = 2.54 cm 1 m $\approx$ 3.28 ft 1 ft = 12 in 1 mi = 5280 ft 1 mi $\approx$ 1.61 km
 1 km = 1000 m 1 m = 100 cm 1 cm = 10 mm 1 kg = 1000 g 1 gal $\approx$ 3.785 L
 1 h = 3600 s 1 min = 60 s 1 yr $\approx$ 3.156×10^7 s

1. Write down the number of significant digits in each measurement.

(a) 48.5 m

(c) 9.58 s

(e) 0.01070 kg

(b) 0.0078 m

(d) 250.0 g

(f) 1.602×10^{-19} C

2. Round each value to three significant figures. Copy the calculator display followed by three dots, then round: $\sqrt{2} = 1.4142135 \dots \approx 1.41$

(a) $\sqrt{7}$

(b) The Earth–Moon distance,
384,400 km

3. Write in scientific notation, rounded to three significant figures.

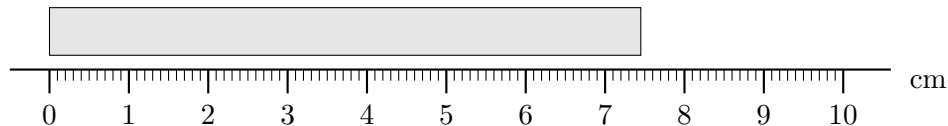
(a) The speed of light, 299,792,458 m/s

(b) The wavelength of green light,
0.000000532 m

4. Write each value as an ordinary decimal number.

(a) The radius of the Earth, 6.37×10^6 m (b) A GPS signal delay, 6.7×10^{-2} s

5. The bar below is measured with a centimeter ruler marked in millimeters.



- (a) Record the length of the bar. Include one estimated digit — the last digit, the one you are not certain of.
- (b) How many significant figures does your measurement have?
- (c) Write the same length in millimeters, and again in meters.
6. Convert between units. Write each conversion as a fraction and show the units cancelling.
- (a) One World Trade Center is 1776 feet tall. What is its height in meters?
- (b) A US nickel has a mass of exactly 5.000 grams. What is the mass of a roll of 40 nickels, in kilograms?
7. The marathon world record is 2 h 00 min 35 s for a distance of 42.195 km. Find the runner's average speed in meters per second, and again in kilometers per hour.

8. A Frecciarossa train runs between Rome and Milan at 300 kilometers per hour.
- (a) What is this speed in meters per second?

 - (b) What is this speed in miles per hour?
9. A badminton smash has been measured at 493 kilometers per hour — the fastest projectile in any racket sport. A baseball fastball leaves the pitcher's hand at 95 miles per hour.
- (a) Convert both speeds to meters per second.

 - (b) The pitcher's mound is 60.5 feet from home plate. How long does the fastball take to reach the plate? A typical human reaction time is 0.25 s — what does your answer say about hitting a fastball?
10. The mass of a proton is 1.673×10^{-27} kg and the mass of an electron is 9.109×10^{-31} kg. How many times more massive is the proton? Give your answer with the correct number of significant figures, and say why that is the right number.

11. The Atlantic Ocean is widening at about 2.5 centimeters per year — roughly the rate your fingernails grow.
 - (a) Express this rate in meters per second, in scientific notation.
 - (b) How many years will it take the Atlantic to widen by 1.0 meter?
12. A student measures the length of the hallway three times with a 30.0 m tape and records 24.1 m, 24.0 m, and 24.2 m.
 - (a) What is the average of the three measurements?
 - (b) The student's calculator shows 24.1000000, and the student writes 24.1000 m in the lab notebook. Explain what is wrong with that.
13. Challenge: Gasoline in Rome costs €1.82 per liter. In New York it costs \$3.45 per gallon. On the day you check, €1 buys \$1.09.
 - (a) What is the Rome price in dollars per gallon?
 - (b) How much more does a gallon cost in Rome than in New York?
 - (c) A Fiat's tank holds 45 liters. What does it cost, in dollars, to fill it in Rome?